

Comparing Hemorrhage in Human Physiology Simulation Tools: How They Compare With Expected Human Physiology and Each Other

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Introduction: Several different whole-body physiology simulation tools (PST) using modeling techniques are now available with potential use for healthcare simulation, but these novel technologies lack objective analysis from an independent organization.

Methods: We identified BioGears, HumMod, and Muse as 3 PSTs that met our requirements for testing. We ran mild, moderate, and severe hemorrhage scenarios on each PST and collected outputs for comparison with each other and published human physiology data.

Results: All PSTs tested followed the expected tachycardic and hypotensive response to hemorrhage for all levels of severity with variable qualitative patterns. Complete data for analysis were not available in all PSTs for urine output, stroke volume, blood volume, hemoglobin, and serum epinephrine concentration, but the partial findings are discussed in detail. We determined the predicted time to reach hemorrhage shock based on the hemorrhage guidelines and compared this with time to cardiovascular collapse from each PST. Overall, the differences from known human physiology were much larger than expected before testing and trends show HumMod with the smallest difference for severe (−6.25%) and moderate (−1.42%) and Muse with the smallest difference for mild hemorrhage (27.9%). BioGears demonstrated the largest differences in all classifications of severity.

Conclusions: Our analysis of currently available whole-body PSTs provides insight into the novel, evolving field. We hope our efforts shed light to a wider audience to the exciting developments and uses of mathematical modeling for whole-body simulation and the potential for integration into healthcare simulation for medical education.

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Key Words: Healthcare simulation, hemorrhage, physiology.

The modeling of complex physiologic processes, which was once isolated within the field of systems biology, is now expanding into clinical translational fields, such as medical education.^{1,2} Advances in physiologic modeling and incorporation into physiology simulation tools (PSTs) have the potential to further transform healthcare simulation.³

Healthcare simulation has seen progress in anatomic components, such as improving the fidelity of task trainers, and in overall advancement of whole-body simulation tools. However, currently available commercial whole-body manikins are still with limitations. For example, some rely on a human technician

to observe the behavior of simulation participants and manually change parameters, such as vital signs, following a predetermined script, or make a judgment call whether a treatment was sufficient to move in a certain direction on a simple algorithm for a certain physiologic condition.

A novel approach to medical simulation is the adaptation of advanced mathematical simulation and modeling, seen in systems biology, into educational tools for healthcare simulation training. This technology, in comparison with the simple algorithmic and human technician-based systems, takes hundreds to thousands of parameters into consideration when calculating the current physiologic state in real time, similar to our understanding of the physiologic processes in the human body. Several different whole-body PSTs using modeling techniques are now available, but these novel technologies lack objective analysis from an independent organization.

Our aim is to formulate standardized testing protocols (by prespecifying hemorrhage rate, location, output data to be collected, and simulated patient sex, height, and weight) for experimentation on select PSTs to collect objective data for fair comparison and evaluation of physiologic fidelity of the systems. We identified 3 software products with the goal of providing whole-body physiology simulations for our testing purposes and ran hemorrhage scenarios to collect data for an objective analysis comparing the fidelity of these PSTs hemorrhage simulation by comparing their output with independently gathered human physiology hemorrhage data available in the literature.

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METHODS

Simulation Tools

We defined a whole-body PST as software using mathematical models to simulate the integrated biologic functions of human organ systems, which are capable of changing physiology in response to external inputs. We identified 3 PSTs that met these requirements and were accessible for experimentation: BioGears (Applied Research Associates, Albuquerque, NM) HumMod (HC Simulation, Canton, MS), and Muse (CAE Inc, Montreal, Canada).

Each PST applied unique mathematical algorithms to model whole-body physiology and validated their models with published human physiology data, when available. BioGears is an open-source whole-body PST based on modeling of the circulatory system after electrical circuit analogs with physics-based lumped parameter models for feedback mechanisms with conservation of mass and momentum.⁴⁻⁶ HumMod is an integrative, deterministic model of human physiology using approximately 10,000 interconnected algebraic and differential equations to represent physiologic, anatomical, or biochemical relationships based on human physiology experimental data.⁷⁻⁹ Muse is a software that integrates numerous proprietary physiologic models developed by Medical Education Technologies, Inc (METI), now CAE Inc, specifically for healthcare simulation.

Physiology

This article focuses on hemorrhage testing because of the availability of comparative human physiology literature, relevance to current healthcare simulation practice, and the modeling availability using the selected PSTs. Before testing, we selected input and output parameters for scenarios and developed definitions for when to cease scenarios, which were applied uniformly across the PSTs. The quantitative definitions and further details used for testing are available in Table 1.

For inputs, mild, moderate, and severe hemorrhage from the left leg was evaluated. The severity scale was calibrated to Advanced Trauma Life Support (ATLS) guidelines of 50, 100, and 500 mL/min for mild, moderate, and severe hemorrhage, respectively.¹⁰ Experiments were carried out using a standardized patient profile (33-year-old man, 175 cm, 70 kg), which was representative of our target simulation profile and could be closely approximated in all 3 selected PSTs.

Discrete physiologic outputs were chosen for comparison. We selected output parameters based on human physiology comparative data and physiologic values that would be desirable in a simulation training scenario as determined by our clinical and simulation scientist experts. We elected to collect heart

rate (HR, beats per minute), mean arterial pressure (MAP, millimeter of mercury), blood volume (BV, milliliter), cardiac output (CO, milliliter per minute), stroke volume (SV, milliliter), hemoglobin (Hgb) content (gram), respiratory rate (RR, breaths per minute), urine output (OUP, milliliter per hour), and epinephrine concentration (picogram per milliliter).

We developed hemorrhage scenarios running at a specified severity for an indefinite amount of time until modeling reached an end point of cardiovascular collapse or until the scenario reached 30 minutes of total run time to improve the feasibility of testing and reduce the generation of noninformative output data.

Comparison Data

Before data collection, we researched existing literature on qualitative and quantitative metrics for expected human physiology for hemorrhage. For comparison with human physiology data, we used the American College of Surgeons ATLS hemorrhage classification and supporting documentation¹¹ (Table 2). We used the ATLS blood loss formula to calculate the time to hemorrhagic shock and compare this with the time of cardiovascular decompensation in the PSTs.

Data Collection and Analysis

Quantitative data were generated by following protocols for inputting physiologic states into each PST using a graphical user interface or XML file creation and collecting output data. The prespecified output parameters of interest were collected every 5 seconds for each experimental run on each PST, which produced output data in comma separated value files. We imported the output data into database software using the software program R to create long-form spreadsheets combining data from each physiology engine in columns along time in rows for statistical analysis, graphical representation, and comparison with established human physiology.

For data analysis, we graphed each parameter with data from each PST (depending on the availability of useable output data for each given parameter) and visually analyzed the data and described the graphical trends. We also compared the time with cardiovascular collapse for the PSTs in the various severities to the calculated time to severe hemorrhagic shock based on ATLS human physiology data using a percent variation calculation.

Quality Assurance and Reproducibility of Data

For quality assurance, we contacted representatives from each of the PSTs and provided our protocol, input, and output data for their appraisal. All 3 entities provided feedback and agreed with the technical aspects of our experimentation and effective use of their products. We created our testing protocols before contacting any physiology simulation company and did not change our initial protocols based on their input.

We aimed to obtain confidence that the PSTs could reliably reproduce the same output data and our findings could be validated. We accomplished this by rerunning one physiology scenario (severe hemorrhage ran indefinitely) using the same software version of each PST and calculating basic statistical information comparing the 2 data sets obtained for each

TABLE 1. Protocols, Parameters, and Severity Levels for Hemorrhage

Hemorrhage Scenarios	
Protocol	Run 3 severities through (a) until parameters stable for 2 min, (b) 30 min of total time, or (c) end point reached
Parameters	HR (bpm), MAP (mm Hg), BV (mL), CO (mL/min), SV (mL), Hgb content (g), RR (bpm), epinephrine serum concentration (pg/mL), and UOP (mL/min)
Severities	Mild: 50-mL/min hemorrhage rate Moderate: 100 mL/min Severe: 500 mL/min

TABLE 2. The ATLS Classification of Hemorrhagic Shock

	I	II	III	IV
Blood loss, mL	<750	750–1500	1500–2000	>2000
Blood loss, % BV	<15%	15%–30%	30%–40%	>40%
Pulse rate, per min	<100	>100	>120	>140
Blood pressure	Normal	Normal	Decreased (SBP < 90 mm Hg)	Decreased (SBP < 80 mm Hg)
RR, per min	14–20	20–30	30–40	>35
UOP	>30	20–30	<20 mL/hour	Negligible

Adapted from *Advanced Trauma Life Support (ATLS) Student Course Manual*. 9th ed. Chicago: American College of Surgeons, 2012:69.
SBP indicates systolic blood.

PST. We calculated the average, maximum value, minimum value, standard deviation, and variance of the values and used them for comparison between the 2 runs of the same physiology scenario.

RESULTS

Reproducibility

For BioGears and HumMod, the mean and range were identical for all out values with zero values for standard deviation and variance (Table 3). The 2 runs gave identical output data for selected parameters and the engine behaves in a deterministic fashion. For Muse, repeat testing did not produce 100% concordance between runs, but the data were very similar. The range of values for all the parameters were identical (except the maximum Hgb, which was 22.5 and 22.7 g/dL for the 2 runs) and the mean values for all parameters had an average standard deviation of 0.189 indicating only a very small difference was found between the 2 runs. Muse does not specify how their engine calculates output data and it is unclear why the output data were not 100% concordant. It is clear that the engine is not completely deterministic from the data; however, it is not clear whether the variance is probabilistic or representative of human physiology variability. Our reproducibility testing gave us confidence that our findings could be replicated and validated.

Heart Rate

Heart rate data in beats per minute from the 3 PSTs running indefinitely (until the virtual patient reached an irreversible state) are graphed for all 3 PSTs, side-by-side, to demonstrate how they handle the severe (500 mL/min) hemorrhage scenario (Fig. 1). All engines followed the expected tachycardic response to hemorrhage.^{12,13} HumMod demonstrates a worsening

tachycardia, followed by a precipitous drop in HR. After this, there is a short, subsequent rise in HR and sharp drop after this just before cardiovascular collapse that we cannot explain; however, this could be related to irregularities in rhythm seen with cardiac ischemia. Muse shows increasing tachycardia that plateaus for a period and then develops a variable tachycardia and bradycardia for a period before full cardiovascular arrest. The Muse model has a “cardiac ischemia index” feature that promotes cardiac arrhythmias as the heart develops ischemia (seen at the 450 second mark in Fig. 1), which could explain this finding in our HR testing data. BioGears becomes more tachycardic with no variability just before cardiovascular collapse.

Mean Arterial Pressure

Mean arterial pressure in millimeter of mercury was collected and available for analysis in all 3 PSTs. The expected physiology is a gradual decrease in MAP over time as blood loss accumulates, followed by a steeper decrease in MAP near decompensation around the time of cardiovascular collapse. The shallower initial decline in MAP is due to the human physiologic compensation for decreased BV by increased cardiac rate, contractility, and peripheral vascular resistance.^{13,14} All 3 PSTs demonstrated a decline in MAP over time as blood loss accumulated in our moderate hemorrhage testing (Fig. 2). HumMod showed a steady decline in MAP over time, starting at 95 mm Hg and then reaching a stabilization at approximately 80 mm Hg, followed by a steep decline to 76 mm Hg at the time of cardiovascular collapse. BioGears showed a clear, linear decline in MAP over time with small spikes and dips above and below the linear trend line, starting at 94.8 mm Hg and ending at 68.3 mm Hg. Muse showed a multiphasic decline in MAP over time with a larger range, at first with a

TABLE 3. Muse Reproducibility Testing Data

	HR	HR-2	MAP	MAP-2	RR	RR-2	CO	CO-2	Hgb	Hgb-2
Mean	108.2745098	108.6078	38.04118	38.84412	8.892157	8.794118	2.257843	2.218627	15.8049	15.87059
Max	147	147	93.6	93.6	11	11	5.2	5.2	22.5	22.7
Min	64	64	−3.3	−3.3	0	0	0	0	14.4	14.4
STD	20.4170172	20.77897	36.31331	36.43822	3.014509	3.128718	1.872967	1.861689	2.374384	2.446966
Var	416.8545913	431.7655	1318.657	1327.744	9.087265	9.788876	3.508007	3.465887	5.637699	5.987641
Mean Var	0.055556		0.055556		0.004806		0.000769		0.002157	
Mean STD	0.235702		0.567765		0.069324		0.02773		0.046447	

Repeat testing data for muse are displayed.

Mean STD = 0.1893936.

Data from BioGears and HumMod is not shown because of 100% concordance between runs (all numbers were equal between runs and no statistical analysis could be performed). Output parameters are labeled along the columns with “-2” indicated data from the second run. Statistical parameters are labeled along the rows. Note the close proximity of the means that all minimum and maximum values are identical aside from the maximum Hgb, and the small variation and standard deviation between the data sets. Abbreviations: max, maximum value; min, minimum value; STD, standard deviation; Var, variance.

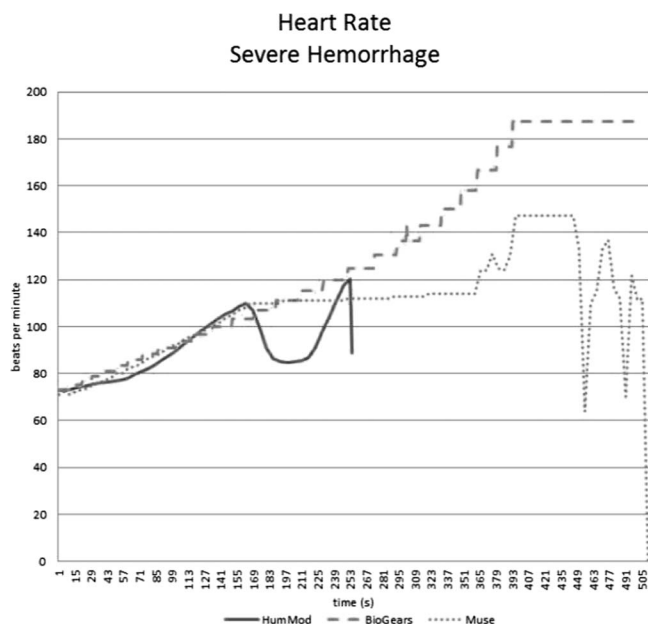


FIGURE 1. Severe hemorrhage graph comparing all 3 PSTs. The HR (beats per minute) calculated from severe hemorrhage ran indefinitely on BioGears, HumMod, and Muse are graphed across time (second) for comparison purposes. All 3 PSTs show tachycardia with eventual cardiac collapse at different times. HumMod reached an irreversible state at 225 seconds, much earlier than BioGears (497 seconds) and Muse (511 seconds). The expected physiology predicts that hemorrhage shock will occur when 2000 mL of blood is lost, which would occur at 240 seconds.

shallower decrease in MAP from the starting point of 91.6 to 71.8 mm Hg, then a steep decline from 71.8 to 32.9 mm Hg, a short plateau, followed by a steep, jagged appearing decline ending at 0 mm Hg.

Urine Output

Urine output in milliliter per minute was collected and available for analysis in HumMod and was not available as an output for Muse, and the data collected from BioGears were not usable for analysis after collection. The expected human physiology during hemorrhage is to have decrease in UOP

over time as blood loss accumulates. It is thought that the human body shunts blood flow away from the kidneys to avoid diuresis and direct blood to more critical tissues during times of blood loss.¹¹ In moderate hemorrhage testing, HumMod demonstrated a pattern of decrease in UOP over time of hemorrhage, which followed a pattern of steep decline from the starting rate of 0.7 mL/min with the onset of blood loss and then a flattening out of the rate of decline to a steady, low state at 0.09 mL/min.

Respiratory Rate

The RR in breaths per minute was collected and available for analysis in all 3 PSTs. The expected physiology is no change to a slight increase in RR. Historical physiology data demonstrate an increase in RR in the middle to late stages of hemorrhage, as a result of decreased CO passing through the pulmonary vasculature for gas exchange, in line with Fick principle, which requires an increase in RR to compensate for this change.^{11,15} However, in a more recent systematic review on hemorrhagic shock human physiology data, tachypnea was found to be an inconsistent finding, even in the late stages of hemorrhagic shock.¹⁰ In our moderate hemorrhage testing, HumMod showed an increase in RR until nearing hemodynamic collapse, when the RR went from 16 to 40 breaths per minute in the final few seconds. BioGears showed a steady, small decrease in RR as hemorrhage proceeded, starting at 16 breaths per minute and ending at 12.5 breaths per minute before reaching hemodynamic collapse, when the RR shot up to 16.75 breaths per minute. Muse demonstrated a flat RR throughout the hemorrhage scenario, with a base rate of 11 breaths per minute before dropping to 9 breaths per minute around the time of hemodynamic collapse (Fig. 3).

Cardiac Output

The cardiac output in liter per minute was collected and available for analysis on all 3 PSTs. The expected physiology is an initial decrease in CO, followed by an increase due to physiologic compensation with changes to vascular tone and HR, and then further decline as decompensation occurs.¹³ In our moderate hemorrhage testing, HumMod showed a sigmoid

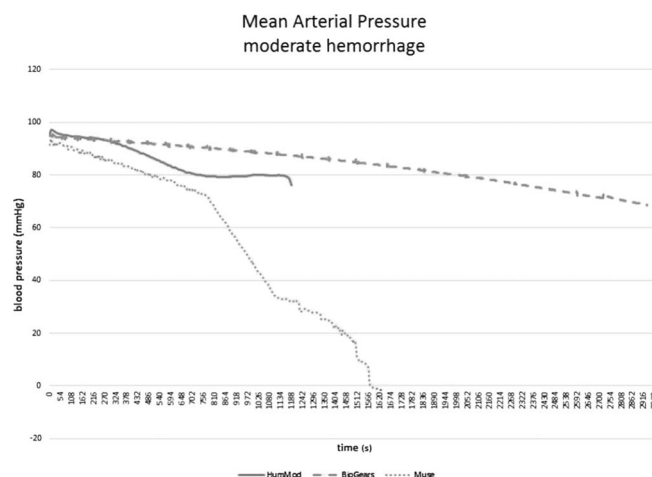


FIGURE 2. Mean arterial pressure during moderate hemorrhage graph comparing all 3 PSTs. The MAP is graphed along the y-axis in millimeter of mercury from the data collected for moderate hemorrhage ran indefinitely on BioGears, HumMod, and Muse are graphed across time (second) for comparison purposes. All 3 PSTs demonstrate the expected hypotension with variable slopes and variation. Cardiovascular collapse was reached at 2935, 1630, and 1183 seconds for BioGears, Muse, and HumMod, respectively.

shaped curve of CO over time, with an initial decrease, followed by an increase to peak, then decrease to nadir, and a final uptick at the time of cardiovascular collapse with a range of 5.2 to 3.1 L/min. BioGears showed a near linear decrease in CO over time; however, the value had small, frequent upward and downward spikes throughout. The range of BioGears CO was 5.6 to 2.9 L/min. Muse showed a consistent decline throughout the hemorrhage experiment from 5.2 to 0.1 L/min. Of note, HumMod reached cardiovascular collapse before BioGears and Muse, which continued to have further decline in CO after HumMod had stopped running (Fig. 4).

Stroke Volume

The SV was collected and available in HumMod and BioGears, but not available in Muse. The expected physiology is a decrease in SV over time.¹³ Both HumMod and BioGears showed a clear, near linear, decline in SV as moderate hemorrhage continued in the scenario (Fig. 5). For HumMod, SV started at 71.4 mL and fell to 42.7 mL, whereas SV in BioGears started at 76.8 mL and fell to 16.39 mL. Of note, HumMod and BioGears had similar starting points and rates of decline in SV; however, BioGears continued to run after HumMod had already reached an irreversible state, which explains the ending SV discrepancy.

Blood Volume

The BV in milliliter was collected and available for analysis in HumMod and BioGears, but not available in Muse. The expected physiology is accumulation of blood loss over time from hemorrhage resulting in lower BVs.¹⁶ In our moderate hemorrhage scenario, this expected physiology was clearly demonstrated by HumMod with a clear, near linear decline in blood loss over time, with HumMod starting at 5406 mL and dropping to 3828 mL, whereas BioGears started at 5064 mL and dropped to 2500 mL. Of note, HumMod and BioGears had similar starting points and rates of decline in BV; however, BioGears continued to run after HumMod had already reached an irreversible state, which explains the ending BV discrepancy.

Hemoglobin

The Hgb in gram per deciliter was collected and available for analysis in HumMod and Muse. The values obtained from BioGears were nonsensical and may have been collected erroneously during the data collection process. The expected human physiology is a stable Hgb, with or without a small decrease as time proceeds.¹⁶ This is due to loss of whole blood, which should not affect the Hgb as it is a value of the proportion of Hgb in the existing blood, which should not be changed by loss of whole blood. However, a small decline would be expected over time as human physiology attempts to compensate for blood loss by volume redistribution, which would pull non-Hgb containing fluid into the blood, resulting in dilution of the existing Hgb content. In moderate hemorrhage testing, HumMod and Muse both demonstrated the expected trend of stable Hgb during the hemorrhage scenario. HumMod started at 15 g/dL and ended at 13.7, showing a slight decrease in Hgb content. Muse started at 14.4 g/dL and ended at 14.7 g/dL, showing a stable or minimally increased Hgb.

Epinephrine Concentration

The epinephrine concentration in picogram per milliliter was collected in HumMod and BioGears, but not available in Muse. The HumMod data were available for analysis, whereas the BioGears data collected contained values too small to use for analysis and to detect meaningful change. The expected physiology is to have increasing epinephrine levels over time as blood loss accumulates.¹⁷ Epinephrine release is one of the physiologic mechanisms to maintain blood pressure and adequate tissue perfusion, which acts by increasing vascular tone and cardiac contractility to compensate for increasing blood loss. In moderate hemorrhage testing, HumMod demonstrated a clear, steep rise in epinephrine concentration in the serum over time in the hemorrhage scenario.

Time to Hemorrhagic Shock/Cardiovascular Collapse

To gather more quantitative comparisons with the PSTs and expected human physiology, we determined the predicted time to reach hemorrhagic shock based on the ATLS guidelines

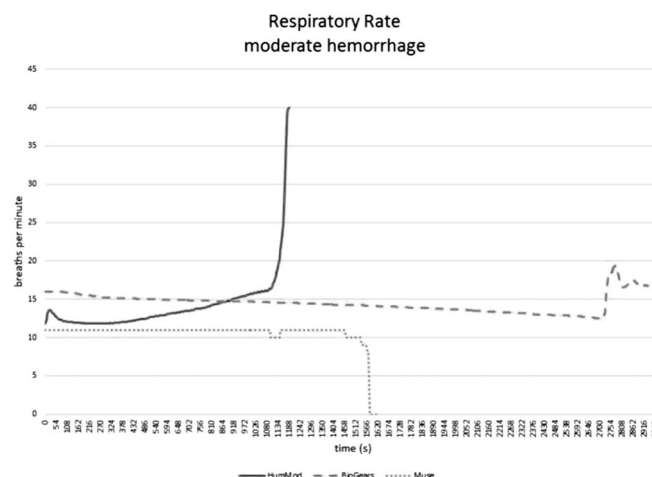


FIGURE 3. Respiratory rate during moderate hemorrhage graph comparing all 3 PSTs. The RR is graphed along the y-axis in breaths per minute from the data collected for moderate hemorrhage ran indefinitely on BioGears, HumMod, and Muse are graphed across time (seconds) for comparison purposes. HumMod shows an increasing respiratory rate with a late, extreme tachypnea response, BioGears shows a decreasing RR until cardiovascular collapse when the rate spikes, and Muse shows a stable RR with a decrease to zero at the time of cardiovascular collapse. Cardiovascular collapse was reached at 2935, 1630, and 1183 seconds for BioGears, Muse, and HumMod, respectively.

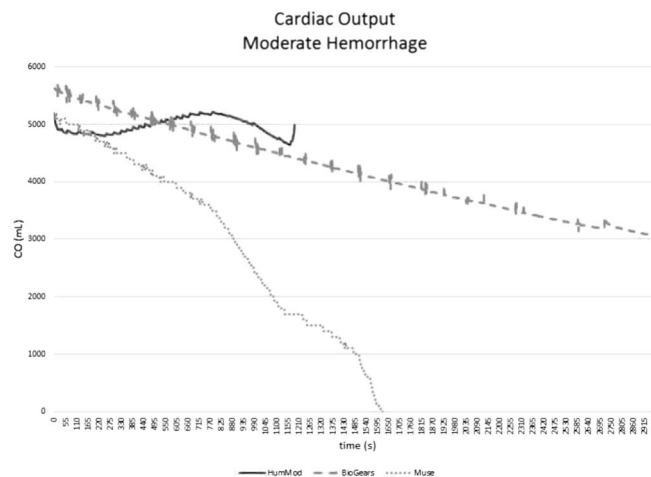


FIGURE 4. Cardiac output during moderate hemorrhage graph comparing all 3 PSTs. The CO is graphed along the y-axis in milliliter from the data collected for moderate hemorrhage ran indefinitely on BioGears, HumMod, and Muse are graphed across time (seconds) for comparison purposes. HumMod shows a sigmoid shape pattern, whereas BioGears and Muse show declining CO over time with varying slopes. BioGears data show a “tram track” appearance with the small, frequent spikes. Cardiovascular collapse was reached at 2935, 163, and 1183 seconds for BioGears, Muse, and HumMod, respectively.

(ex. it would take 240 seconds to lose 2000 mL of blood at 500 mL/min, resulting in late-stage hemorrhagic shock) and compared this the data we gathered from the tools. HumMod reached an irreversible state at 225 seconds, much earlier than BioGears (497 seconds) and Muse (511 seconds). Table 4 shows the time to hemorrhagic shock and cardiovascular collapse from hemorrhage scenarios across the 3 PSTs compared with the expected time to hemorrhagic shock and cardiovascular collapse based on calculations from ATLS data from human physiology. In this example, the trends of hemorrhagic shock from our output parameters, which are displayed graphically in Figures 1 to 5, closely followed expected physiology; however, the time to cardiovascular collapse, which are listed in Table 4, was highly variable and inconsistent with known human physiology. The difference in time to cardiovascular collapse from established human physiology data was more than 100% on 4 of the 9 test runs, with BioGears having more than 100% difference on all runs (107%–145%), HumMod having a range of –6.25% to 77.4% difference, and Muse

having a range of 27.9% to 113%. Overall, the differences from known human physiology were much larger than expected before testing and trends show HumMod with the smallest difference for severe (–6.25%) and moderate (–1.42%) and Muse with the smallest difference for mild hemorrhage (27.9%). BioGears demonstrated the largest differences in all classifications of severity.

DISCUSSION

Our analysis of currently available whole-body PSTs provides insight into the novel, evolving field of whole-body physiology simulation and its use for medical education and healthcare simulation. We hope our efforts shed light to a wider audience to the exciting developments and uses of mathematical modeling for whole-body simulation and the potential for integration into healthcare simulation for medical education.

We developed a structured manner for evaluating whole-body PSTs by creating uniform, predetermined testing protocols of various rates of hemorrhage and collecting relevant

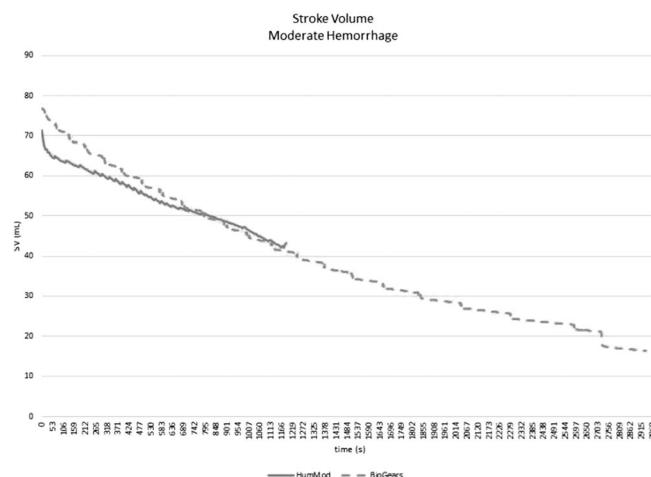


FIGURE 5. Stroke volume during moderate hemorrhage graph comparing all 3 PSTs. The SV is graphed along the y-axis in milliliter from the data collected for moderate hemorrhage ran indefinitely on BioGears and HumMod are graphed across time (seconds) for comparison purposes. Muse did not have SV as an available output. HumMod and BioGears show similar SV values over time with similar rates of decline. Of note, BioGears ran until cardiovascular collapse at 2926 seconds, much longer than HumMod's cardiovascular collapse at 1230 seconds.

TABLE 4. Comparing Time to Cardiovascular Collapse

Source	Time to cardiovascular collapse, s	% difference from human physiology
Severe hemorrhage (500 mL/min)		
Human physiology	240	N/A
BioGears	497	107%
HumMod	225	-6.25%
Muse	511	113%
Moderate hemorrhage (100 mL/min)		
Human physiology	1200	N/A
BioGears	2935	145%
HumMod	1183	-1.42%
Muse	1630	35.8%
Mild hemorrhage (50 mL/min)		
Human physiology	2400	N/A
BioGears	5884	145%
HumMod	4257	77.4%
Muse	3069	27.9%

The calculated time to Class IV hemorrhagic shock (2000 mL of blood loss) is presented as "human physiology" compared with output data until cardiovascular collapse in BioGears, HumMod, and Muse simulation tools for severe, moderate, and mild hemorrhage testing runs.

Abbreviation: N/A, not available.

physiology output metrics to clinical practice for comparison with human physiology data and other PSTs. The process of creating standardized protocols that could be used for multiple human physiology stimulation tools before any data collection was paramount to the success of our investigation. This allowed us to objectively collect data in a finite manner to allow for unbiased comparisons. We attempted to create protocols that demonstrated the functionality of the tools without catering to each tool's unique and intricate methods for input and without manipulation of parameters to create the desired response.

Simulation scientists and medical doctors worked together to identify 3 whole-body simulation tools for evaluation and used standardized protocols to ensure uniform testing that generated consistent, objective, quantifiable, and comparable data for analysis. Investigators compared the generated hemorrhage data from the simulation tools to published human physiology data to assess the model's fidelity to known physiology within set parameters. The protocolized testing uncovered both the fidelity and limitations of novel computer human PSTs compared with documented human physiology and accepted clinical guidelines in hemorrhage scenarios.

Overall, all 3 PSTs demonstrated appropriate trends of tachycardia and hypotension for hemorrhage with scalability of response based on the severity of the hemorrhage. The 3 tools did show some variability in the magnitude of these responses, which was more difficult to quantify and compare objectively to human physiology data. In addition, the difference in time to cardiovascular collapse from the PSTs was greater than expected, which could be explained by errors in the modeling of the engines or by the limitations of our choice of comparative human physiology end point for cardiovascular collapse.

HumMod demonstrated the closest correlation with length of time until cardiovascular collapse in comparison with our calculations from ATLS data and would be best suited for medical education use because of this as it would be beneficial to have learners gain familiarity with the time course of mild, moderate, and severe hemorrhage. BioGears, Muse, and HumMod all

did a sufficient job demonstrating the expected major physiologic patterns, such as tachycardia and hypotension, making them suitable for use in healthcare simulation with the caveat that the educator would have to monitor timing and total blood loss if attempting to teach the principal of time until late-stage hemorrhagic shock. Muse stood out from the other PSTs as most accurately portraying the trends of tachycardia and hypotension in the expected magnitude during hemorrhage, and if an educator is tracking timing, this would be the ideal PST tested for basic hemorrhagic scenario use in healthcare simulation.

One limitation of our experiment is the human PSTs themselves. We have championed the ability of these tools and described them as the future of healthcare simulation; however, the subfield of computational healthcare simulation is in its infancy and this comes with challenges and limitations. The tools themselves were challenging to operate, requiring extensive training and time familiarizing with basic computer programming techniques to successfully create and run scenarios, which is a significant barrier to bringing the technology into use for healthcare education as this is largely overseen by clinicians rather than computer scientists. The tools used also frequently developed unexpected errors and some underwent major developmental changes during and after our testing, which may render the experiential data collected no longer reflecting the accuracy and functionality of the technologies. Overall, the tools we tested, which we believe are a good representation of the current state of the technology, were challenging to operate for clinicians and unstable from a software perspective with evolving and changing functionality. The issues with the time of cardiovascular collapse being far different than expected were shared with the respected companies of each PST, and the PSTs utility for use within healthcare simulation would be improved if these issues were addressed.

Another major limitation of our work is the lack of criterion standard human physiology data on late-stage hemorrhage. We elected to use the ATLS guidelines for expected human physiology, as these guidelines are familiar to a large number of clinicians and accepted within the healthcare community. The ATLS guidelines provide a reference point for comparison purposes, but they describe categories rather than a continuum of ongoing hemorrhage, like we tested in our scenarios. Because of this, one must make the assumption that continuous hemorrhage would follow the ATLS guideline categories, which may or may not be the case.

In the future, the protocolized methods for testing whole-body PSTs will allow for continued testing of new and refined technologies with the ability to compare across and within software platforms. This streamlines the testing process and removes some of the subjectivity with evaluation of available products. We believe that the lasting benefit of this research effort will be ongoing and future application of analysis of PSTs rather than a snapshot of currently available PSTs ability to run hemorrhage scenarios.

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